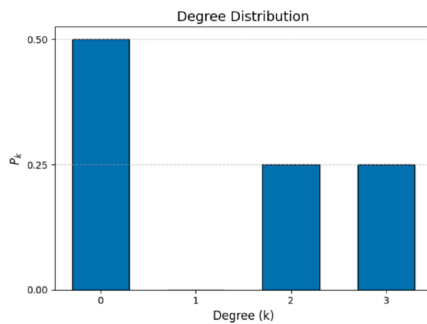


Lesson-3: Quiz (graded)

1

Multiple choice 1 point

What would be the average degree and the standard deviation of the degree distribution for a graph with 8 nodes whose empirical degree distribution is as shown below?

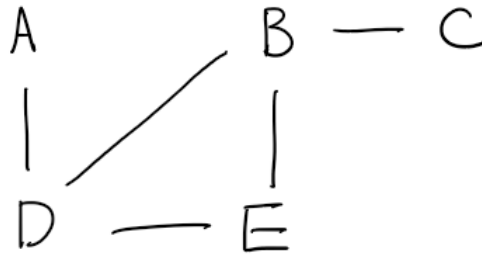


- ☐ $k = 5/4$, $\sigma = \sqrt{27/4}$
- ☐ $k = 5/4$, $\sigma = \sqrt{23/4}$
- ☐ $k = 7/4$, $\sigma = \sqrt{23/4}$
- ☐ $k = 7/4$, $\sigma = \sqrt{27/4}$

2

Multiple choice 1 point

What is the expected degree of a node's neighbor in the graph below?

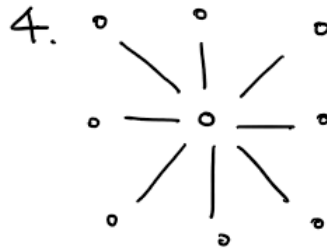
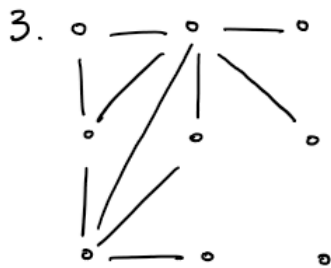
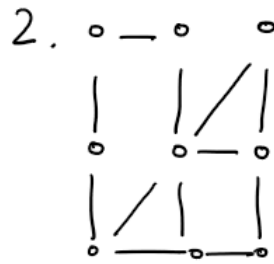
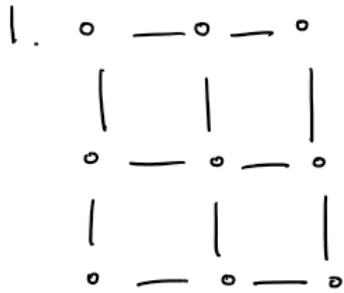


- ☐ 2.0
- ☐ 2.4
- ☐ 2.5
- ☐ 2.6

3

Multiple choice 1 point

Which of the following graphs would be the most extreme instance of the Friendship Paradox?

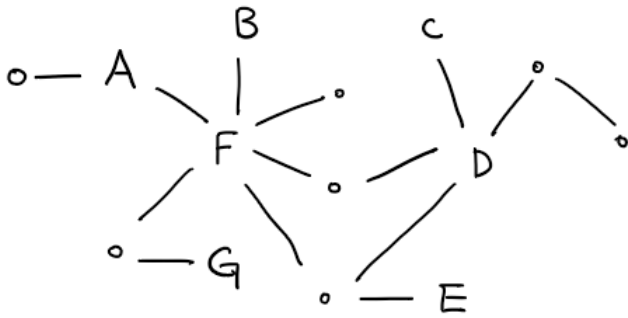


- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

4

Multiple choice 1 point

If you can only vaccinate 2 people, who would you immunize based on the following network of contacts?



- ☐ (A,E)
- ☐ (B,F)
- ☐ Two randomly selected individuals
- ☐ (F,D)

5

Multiple choice 1 point

Consider the $G(n,p)$ model of random graphs with $n=1000$ nodes. What is the minimum value of p so that largest connected component is expected to include all nodes? Round off your answer to three decimal places.

- ☐ 0.006
- ☐ 0.007
- ☐ 0.008
- ☐ 0.009

6

Multiple choice 1 point

Which of the following statements is true for the **in-degree distribution** of random DIRECTED network? The number of nodes is n and the probability that any node x connects to any node y (other than x) is p

1. $p_k = \binom{n}{k} p^k (1-p)^{n-k}$

2. $p_k = \frac{1}{2} \binom{n-1}{k} p^k (1-p)^{n-k-1}$

3. $p_k = \binom{n-1}{k} p^k (1-p)^{n-k-1}$

4. $p_k = 2 \binom{n-1}{k} p^k (1-p)^{n-k-1}$

☐ 1

☐ 2

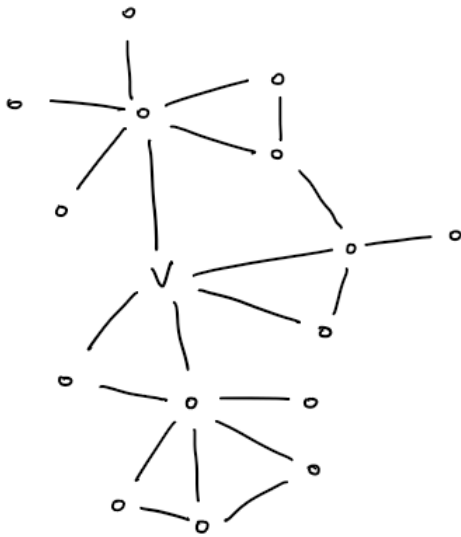
☐ 3

☐ 4

7

Multiple choice 1 point

What is the average neighbor degree for node v ?



- ☐ 4
- ☐ 3
- ☐ 2
- ☐ 1

8

Multiple choice 1 point

Which one of the following statements is **True**?

- ☐ We expect that a $G(n, p)$ network is connected if the average node degree is 1.
- ☐ The friendship paradox can be observed in any network
- ☐ We use Poisson distribution instead of Binomial distribution to model degree distribution of the $G(n, p)$ networks because it is more accurate for small p values.
- ☐ Effects of the Friendship Paradox is more significant if $\frac{\bar{k}^2}{\bar{k}} \gg \bar{k}$